MSML606 Project Proposal


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LegiMap: An open-source tool to search and visualise legal citations and literature survey

Background and Motivation

The U.S. legal system operates on the principle of *stare decisis*, requiring practitioners to identify relevant precedents across millions of court opinions. However, the volume of case law is growing superlinearly, making manual literature surveys a "needle in a haystack" problem. Researchers face a significant cognitive burden when parsing unstructured text to verify "good law," which can lead to oversights. Currently, the industry relies on proprietary citators like LexisNexis and Westlaw. These tools are prohibitively expensive, creating a "paywall to justice" for small firms and independent researchers.

Problem Statement:

In this project, I plan to build **LegiMap** as an open-source search and visualisation tool to facilitate literature surveys in the legal domain. By transforming static citations into a dynamic network of nodes and edges, the platform allows researchers to manually identify seminal rulings and navigate the "shape of the law" through an interactive interface (similar to LitMaps).
To achieve this at scale, I will manually implement the following algorithms:

i. Normalised Hash Table: For O(1) retrieval and deduplication of "noisy" citation strings.

ii. N-ary Tree: To map multi-directional case lineages and perform efficient breadth-first search for network expansion.

iii. Seed-and-Expand Algorithm: A search and discovery algorithm that uses *bibliographic coupling* to suggest relevant cases based on shared references.

Data Overview

Data Source:

I plan to use the **Caselaw Access Project (CAP)** dataset, specifically the bulk exports provided by the Harvard Library Innovation Lab and mirrored on Hugging Face. This dataset encompasses over 6.7 million unique U.S. court cases. To meet the project requirement of 10,000+ items, I plan to filter this corpus to focus on a specific subset, such as **"all** federal supreme court cases (1970–2024)".

Data Structure:

The data is provided in a semi-structured **JSON-L** format, with each record containing critical metadata:

- Unique Identifiers: Citation strings (e.g., `410 U.S. 113`) and internal IDs.
- Temporal Data: `decision_date` for time-scaled mapping.
- Relational Data: `cites_to` fields, which form the edges of our citation network.

Data Preprocessing and Challenges (Expected)

I have identified four potential challenges I may face when working with this dataset. I have described each of them below

i. Legal citations lack a strict, machine-readable standard in raw text. The same landmark case may be referenced using various permutations—such as `410 U.S. 113`, `410 US 113`, or `410 U. S. 113`. This prevents direct string matching and breaks the ability to perform constant-time O(1) lookups in a global index.

ii. Bulk legal datasets often exceed several gigabytes and contain millions of individual JSON records. Loading the entire corpus into memory to build a citation network would cause the system to fail, necessitating a strategy that supports high-volume ingestion without exceeding hardware limits.

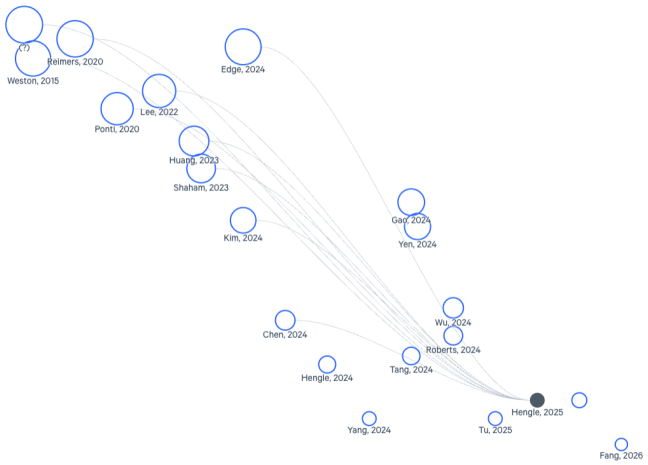
iii. Many open-source legal records are derived from historical physical volumes via Optical Character Recognition (OCR). This results in common character substitutions (e.g., "C0urt" instead of "Court" or "1" appearing as "l") that obscure the legal meaning and prevent the identification of critical keywords.

iv. While some modern datasets provide metadata fields, many citations are buried within thousands of words of judicial prose. Identifying the "hidden" citation network within these unstructured opinions is difficult, as the relationships are not explicitly linked in the raw data files.

Evaluation Metrics

To ensure **LegiMap** meets both computational efficiency and research utility standards, the project will be evaluated using two primary metrics:

- **Hash Table load factor & Collision Rate:** I will monitor the performance of our manual Hash Table. Success is defined by maintaining a load factor that ensures an average **O(1)lookup time** across 10,000+ items, proving that my custom collision-handling logic scales effectively with the dataset.
- **Recall (for discovery):** We will evaluate the "Litmaps" discovery logic by establishing a **Ground Truth** path—a set of seminal cases known to be legally related (for example: *Griswold v. Connecticut* leading to *Roe v. Wade*). Success is measured by the system's ability to achieve a recall@K>85% and to successfully surface these landmark cases within the top visual results using our "Seed-and-Expand" algorithm.

User Interface

The LegiMap UI would look something like this. Given a case ID or description, my search algorithm will find the most relevant metadata and visualise it as a network.